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OCTOBER 17, 1955
SUPERSEDING
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18 January 1954

FEDERAL SPECIFICATION

CHAIRS, STEEL, GENERAL OFFICE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—The steel chairs covered by this specification are a matching or companion group, which are uniform in appearance, comfort, design, and quality of material and represent the Federal standard steel office chairs. They are primarily intended for general offices where there is a need for their use in relation to other standard metal office equipment to maintain uniformity. The type II, class 2 styles are adjustable posture chairs.

1.2 **Classification.**

1.2.1 *Types, classes, and styles.* — The steel chairs shall be of the following types, classes, and styles, as specified in the invitation for bids:

Type I.—Straight (leg) chairs.
  Style A.—With arms.
  Style B.—Without arms.

Type II.—Swivel (rotary) chairs.
  Class 1.—Tilting seat and back.[1]
    Style A.—With arms.
    Style B.—Without arms.
  Class 2.—Non-tilting seat, tilting adjustable back.
    Style A.—With arms.
    Style B.—Without arms.

[1] Fixed back tilts with seat.

1.2.2 *Colors of finish.*—The color of finish shall be as specified in the invitation for bids and shall match the following standard colors adopted by the Federal Supply Service, General Services Administration:

Gray
Olive Green

(Sample panels of the standard colors of finish are obtainable, without charge, from the Business Service Center, Regional Office Building, Federal Supply Service, General Services Administration, Washington 25, D. C., or from the Business Service Center of the nearest GSA Regional Office.)

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

FF-C-81—Casters; Metal Chair.
UU-T-111—Tape; Paper, Gummed (Sealing and Securing).
CCC-A-700—Artificial Leather Upholstery (Synthetic Resin Coated Cloth).
CCC-B-811—Burlap; Jute.
CCC-P-91—Padding; Rubberised Hair or Vegetable Fiber.
LLL-B-631—Boxes; Fiber Corrugated (for domestic shipment).

Military Specifications:

JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
MIL-L-10547—Liners, Case, Waterproof.
MIL-R-17252—Rubber, Cellular (Bonded Shredded).

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Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking for U. S. Military Property.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following publications, of the issues in effect on date of invitation for bids, unless otherwise stated, form a part of this specification.

Commercial Standard:

CS-35-49—Hardwood Plywood.

ASTM Specification:

D-1055-53T—Latex Foam.

(Copies of Commercial Standard may be obtained upon application, accompanied by money order, coupon or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.

(Copies of ASTM Specification may be obtained from American Society for Testing Material, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Steel.* — All steel parts shall be cut, stamped, or die-formed from steel sheets and steel tubing. The steel shall be free from rust, scale, and all other defects. All steel tubing shall be fabricated from the same grade of steel, and shall be seamless or continuously seam-welded. The gage numbers specified for the steel chair parts are the Manufacturers' Standard Gage for Steel Sheet. The decimal equivalent of the steel sheet thickness shown is subject to the established standard thickness ranges (limits).

3.1.2 *Hair padding.*—The hair padding used for chair seats and backs shall conform to type I, class 1, and either grade A or grade B of Federal Specification CCC-P-91 for Padding; Rubberised Curled Hair or Vegetable Fibre. The thickness shall be as specified herein.

3.1.3 *Cotton felting.* — The cotton felting shall be made of unbroken laminated sheets of new garnetted felt at least equal in quality to felt made from a blend of 75 percent U. S. Standard grade 2 low linters and 25 percent U. S. Standard grade low middling cotton 7/8 inch staple. The felt shall weigh not less than 14 ounces per yard 27-inch-width basis, and shall be clean and free from foreign matter. Felt shall be examined before using and approved as specified in 4.3.8.

3.1.4 *Cotton wadding.* — The wadding shall be made of nonabsorbent carded cotton, 35 percent of which shall be not less than 1/2 inch in length. Fifty percent of the cotton may be bleached. Wadding shall weigh not less than 2.8 ounces per square yard.

3.1.5 *Rubber padding.*

3.1.5.1 *Latex foam.*—Unless otherwise specified, the latex foam sheets and molded cushions shall be all natural rubber or a suitable blend of natural and synthetic rubber, and shall conform to the test methods as shown in ASTM Specification designation D-1055, latest issue, as they apply to latex foam. The latex foam sheets shall be solid stock and of the thickness as specified herein and of a compression resistance of ASTM, grade R-32. The molded cored latex foam cushions shall be of ASTM grade R-32 and not less than 3 inches thick.

3.1.5.2 *Bonded shredded foam.*—When specified, the bonded shredded foam for seat cushions

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and back pads shall be made of new synthetic or natural rubber foam that is dry, free from coagulation, excessive flash and other foreign materials, and shall conform to type II, class 2, condition medium, as specified in Military Specification MIL–R–17252, except that each seat cushion shall be molded. The molded shredded foam cushion shall be not less than 3 inches thick before application and the shredded foam for back pads shall be die cut from the specified sheet stock thicknesses before application. Dimensional tolerances shall be as specified in Military Specification MIL–R–17252.

3.1.6 *Cushion springs.*—The cushion springs shall be formed from not less than 0.0990 inch diameter, properly tempered 50 to 70 percent carbon steel spring wire. The springs shall be of a height suitable for the thickness of the cushion specified and shall have not less than 4 coils each. The springs shall not lose more than ⅛ inch in height after six full compressions and shall be stress relieved before using.

3.1.7 *Burlap.*—All burlap used shall conform to Federal Specification CCC–B–811 for Burlap; Jute, and shall weigh not less than 10 ounces per linear yard, 40 inch-width basis.

3.1.8 *Upholstery.* — The upholstery covers shall be of a quality not lower than class 2 of Federal Specification CCC–A–700. The finish shall be that of simulated top-grain leather and the color shall be approximately the color of the chair frame (see 1.2.2).

3.1.9 *Hardware.*

3.1.9.1 *Casters.*—Casters and caster sockets for all type II chairs shall be ball bearing swivel, 2-inch wheel diameter, and wheel composition material, hard rubber composition with soft rubber tread, in accordance with Federal Specification FF–C–81. Plate type casters of the same quality will also be acceptable. Riveted caster wheel axles will be acceptable.

3.1.9.2 *Glides.* — Glides shall be properly nickel, or chromium plated steel or polished stainless steel and shall be of the live-rubber-cushioned and spring steel expansion or rubber-expansion type that can be firmly secured to each leg but may be removed without special tools for replacement. There shall be no metal to metal contact of glide base and chair leg after the glides are applied and the glides shall be not less than 1 1/16 inch diameter with a substantially flat bottom. The base shall be of stamped steel not less than 0.050 inch thick and case hardened after forming to not less than 15 *N* 85 Rockwell or equivalent on comparable scale. The base shall have properly curled edges to contain and retain the rubber cushion which shall fill the inside of the base and extend 3/16 inch above it.

3.1.9.3 *Scuff plates.*—The scuff plates shall be not less than 20-gage (0.0359-inch) corrosion-resisting steel and shall be designed to completely protect the top and end enameled surfaces of the base from abrasions and shall have a developed length of not less than 6 inches. The scuff plates shall have slightly offset or rounded edges closely fitting against the base leg to eliminate sharp exposed edges which might cause injury to the occupant or damage clothing or shoes.

3.1.9.4 *Chair controls.* — The chair control shall provide for height adjustment and revolving of all type II chairs and shall have seat tilt adjustment for type II, class 1, styles A and B chairs, and back tilt adjustment for type II, class 2, styles A and B chairs. All adjustments shall be made by metal knobs or hand wheels that have well formed finger grips and that are easily accessible for all adjustments. The mechanism shall be self locking and positive in height adjustment, and non-binding in revolving, raising or lowering the seat over a range of not less than three inches which shall be accomplished without manually lifting the seat. All mechanisms shall be smoothly finished in the same color as the chair frame and shall be properly lubricated.

3.1.9.4.1 *Chassis, spider-arms and spindle.* —The chassis and spider arms shall be of sufficient strength to pass the required tests specified in 4.3 for type II chairs. The parts shall be assembled by rivets, bolts, or spot-welding, and be designed to give a rigid and properly balanced support between the chair frame and the base at all times. The spindle shall be cold drawn steel not less than 1 inch in diameter and

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shall be welded or pressed fit and riveted to the chassis. An iron casting shall form the chassis into which the spindle shall be pressed fit for a distance of not less than 1¾ inches. A 5/16-inch diameter rivet shall be driven through the casting and spindle, or a formed steel chassis may be used in which the spindle shall be further reinforced by a bracket, tube, or accurately machined part extending around and along the spindle not less than ¾ inch to provide additional strength. The spider arms shall be formed to provide a vertical web of not less than 1 inch and shall be designed for rigidly securing to the seat frame structure at not less than four places with bolts, lockwashers and nuts.

3.1.9.4.2 *Tilting unit.* — Tilting units shall provide the tension against the tilting of the chair seat or back by means of steel springs or rubber torsion units. Tilting tension bolts shall be not less than ⅜-inch diameter, must be entirely supported by the chair control structure, not the seat pan. The bolt shall have ample clearance thru the entire tilting range to prevent breaking wear by friction. The tilting unit shall have a positive safety stop that will limit the tilting action during normal use and in case of failure or breakage of the tension unit. The range of the tilting action shall be not more than 24 degrees for type II, class 1, styles A and B chairs and not more than 17 degrees for type II, class 2, styles A and B chairs. The steel spring tilting mechanism shall consist of either two parallel tempered steel wire spiral springs not less than 3 inches long, nor less than 1½ inches outside diameter, of a wire thickness not less than 0.250 (–0.005) inch or of one steel wire spiral spring not less than 4 inches long, nor less than 1⅞ inches outside diameter, of a wire thickness not less than 0.343 (–0.005) inch.

3.1.10 *Sound deadening material.* — The sound deadening material shall be composed of organic base materials asphaltum impregnated or mineral base materials, which shall be applied to a thickness of not less than 5/32 inch and over not less than 50 percent of the interior area of each leg of the bases. It shall be of such character as to adhere tenaciously without becoming brittle and to resist moisture and insects as well as resist heat of a temperature of 120°F., without running. Alternatively, latex foam or cork blocks not less than 3 inches by 3 inches by 1 inch properly secured into each leg will be acceptable if they provide equal sound deadening effectiveness.

3.1.11 *Plywood for seat and back pan.* — Plywood, if used for seats and backs, shall be constructed in accordance with Commercial Standard CS–35–49 for hardwood plywood, having type II bond, grade 2 face and back veneers, cross banding and cores. The plywood shall be not less than 5-ply and edges not padded shall be smoothly rounded to prevent sharp edges from contacting and damaging upholstery covers.

3.2 Construction.

3.2.1 *Design.*—Figures 1 through 6 illustrate the general exterior appearance, typical construction and minimum dimensions of the chairs covered by this specification. Unless otherwise stated, all dimensions shall have a plus tolerance of 2 inches and a minus tolerance of 0 inch.

3.2.1.1 *Assembly of metal parts.* — The assembly of metal parts shall be by fusion welding or brazing for maximum strength at each joint. Jigs shall be used to insure proper alignment of parts during welding. Where necessary, mechanical joints shall be made with steel rivets, or by cadmium or zinc plated bolts, nuts and lockwashers. Finished chairs shall withstand the tests specified in 4.3.

3.2.1.2 *Welding and brazing.* — All welding shall accomplish rigid joints in proper alignment. Welds shall be sound without porosity and shall be ground smooth to present a finished appearance, and to provide a rigid one piece unit frame that shall be of sufficient strength to meet the prescribed tests, without any parts that will work loose or affect the safety of the occupant during heavy duty daily use, and to withstand rough handling in moving.

3.2.1.3 *Bending of tubes.*—The tubes when bent shall have true and even curves without flattening, crushing or crimping that will affect the appearance or strength of the tubes.

3.2.2 *Seat frame and seat.*

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3.2.2.1 *Seat frame.* — The seat frame shall have gussets or braces welded at each corner and shall be designed for sufficient strength to withstand the tests prescribed in 4.3. The front corners of the seat frame shall be rounded or not less than ¾-inch radius, and all frames shall be set back of seat cushion sufficiently to prevent any uncovered metal of the seat frame from hitting desks or other items of furniture during normal daily use. Provision shall be made for securing the seat frame to the seat by nuts, bolts and lockwashers and to the legs, arms, or chair iron as the type, class or style of chair requires.

3.2.2.2 *Seat.* — Unless otherwise specified, seat shall be not less than 3 inches thick at the edges, full cushioned and shall consist of spring support pan, wadding, springs, burlap, hair pad, cotton and cover. The spring support pan shall be not less than 20-gage (.0359-inch) steel, formed for strength. Alternatively, plywood seat pan not less than ⅜ inch thick as specified in 3.1.11 may be used. Seat pans shall be perforated sufficiently for air ventilation, and shall be covered on the top side with sheet wadding as specified in 3.1.4 to exclude dust and minimise noise. The edge of the steel seat pan shall have a full length rubber bumper strip, or the edges shall be rolled and bumper welted to prevent the edge of the seat pan from damaging the upholstery material during normal use. The springs shall be as specified in 3.1.6 and not less than the minimum number specified for the type, class and style of chair specified. The springs shall be securely fastened together by a border wire and by metal clips or wire spirals forming a complete unit. The springs shall be covered on the top, bottom, and sides with burlap as specified in 3.1.7 and padded on top with a 1½ inch thick pad of grade A rubberized curled hair as specified in 3.1.2 which shall extend down over the top of the spring unit on both sides approximately ½ inch, and 1½ inches on the front. A double layer of cotton felting as specified in 3.1.3 or a 1-inch-thick foam rubber slab shall be placed over the curled hair pad. Loose cotton shall be used to round out the seat for sufficient fullness. The pad, wadding, cotton and springs may be omitted and a full molded latex foam cushion as specified in 3.1.5.1 shall be permanently cemented to the seat pan. The four sides and top shall be covered with the upholstery material specified in 3.1.8 and all seat cushions shall be fully packed and properly crowned. All seams shall be double stitched and the top of the seat shall have welted sides or edges only and the covers shall be securely attached to the underside of the seat pan. The method of attaching the upholstery material shall be completely free from hazardous surfaces or devices and shall prevent slippage and excessive wear of the covers after continual heavy use. When specified, seat cushions shall be of bonded shredded foam as specified in 3.1.5.2 and shall be cemented to the seat pan.

3.2.3 *Back pad.*—Unless otherwise specified, chair back pads for type I, styles A and B and type II, class 1, styles A and B chairs shall consist of a grade B curled hair pad as specified in 3.1.2, ¾-inch thick, over which shall be placed a ½-inch thick layer of soft density foam latex as specified in paragraph 3.1.5.1. Loose cotton may be used to shape and round out the pad. The pad shall be slightly larger than the full size of the back pan and shall present well rounded edges after covering with the upholstery material specified in 3.1.8. The upholstery shall be neatly turned around all edges and secured to the back of the pan. The chair back pads for type II, class 2, styles A and B chairs shall consist of a pad of soft density latex foam rubber not less than ¾ inch thick as specified in 3.1.5.1. When specified, chair back pads shall be of bonded shredded foam as specified in 3.1.5.2 which shall be cemented to the back pan.

3.2.4 *Arm pads.*—The arm pads shall be not less than 1⅜ inches wide, and shall consist of a 1-inch wide strip of 20-gage (.0359-inch) metal, padded with a hair felt pad not less than ¼ inch thick and ½-inch slab of firm latex foam. The pads shall be neatly covered with the same upholstery material as the seat and back, and shall have well rounded and padded edges and corners, and shall be free from sharp protrusions or hard edges which might damage the cover during use, cause injury or catch clothing. The arm pads for fixed back chairs shall cover the top of the arm from a point not to exceed 4 inches from the back of the frame and shall fit around

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the contour of the arm not less than 2 inches below the top of the arm pad support.

3.2.5 *Chafing pad.*—The chafing pad shall be a non-brittle, non-marking material and of a design to prevent the chair back from scuffing or marking walls. The pads shall be securely fastened to the chair back frame and be approximately the color of the upholstery material.

3.3 Detail requirements.

3.3.1 *Type I, style A chairs.* — The type 1, style A chairs shall be of the general design, and not less in size than the minimum dimensions shown in figure 1, and as specified herein. The chairs shall withstand the test as described in 4.3.

3.3.1.1 *Seat.*—The seat frame and seat shall be constructed as specified in 3.2.2 and the seat cushion shall contain not less than 20 springs. The seat cushion shall be securely attached to the seat frame at not less than four places as specified in 3.2.1.1.

3.3.1.2 *Back.* — The back frame, including back legs, shall be formed from one piece of not less than 1-inch by 1-inch 20-gage (0.0359-inch) steel tubing. The back cross brace shall be not less than ⅜-inch by ⅝-inch 22-gage (0.0299-inch) steel tubing, and shall be securely welded in place. The back pan shall be not less than 24-gage (0.0239-inch) steel. Alternatively, a ¼-inch-thick plywood back pan as specified in 3.1.11 may be used. The back pan shall be padded as specified in 3.2.3 and covered on the back with the same upholstery material as the pad with edges folded or rolled to eliminate raw edges of the upholstering material except where edges are neatly concealed by the back frame. The back pan and pad shall be secured to the frame by not less than four screws or in an equally secure manner.

3.3.1.3 *Arm supports.* — The arm support shall be constructed of one piece of not less than 1-inch by 1-inch 20-gage (0.0359-inch) steel tubing which shall be securely welded and brazed on all sides to the back legs and to the seat frame. The arm support shall be completely closed on the lower end but may have a ⅛-inch hole drilled in the closure for finish drainage purposes. The arm pad shall be as specified in 3.2.4 and shall be firmly secured by not less than one rivet-head internally threaded barrel nut and membering flat head machine screw and one key slot and lug.

3.3.1.4 *Legs.*—The front legs shall be die-formed from not less than 20-gage (0.0359-inch) steel with the front corner rounded to an approximate ¾-inch radius at the top and tapering gradually to ⅛-inch radius at the bottom. The front legs shall be not less than 1⅝ inches square at the top and shall taper uniformly to not less than 1-inch square at the bottom. The rear inside corner shall be continuously welded the full length, or it shall be reinforced with a 20-gage (0.0359-inch) steel angle, with flanges not less than ½-inch wide which shall be projection welded to the adjacent flanges of the outer leg with welds spaced not more than one inch apart. All corners of the leg shall be smoothly finished without any visible seams after finishing. The lower end of all legs shall be neatly closed by not less than an 18-gage (0.0478-inch) steel plate securely and smoothly welded in place. The plate shall be perforated with a hole of ⅝-inch diameter to receive and firmly grip the glide specified in 3.1.9.2 and shall provide a positive bearing for the rubber cushion of the glide. The leg braces or stretchers shall be not less than ⅜-inch by ⅝-inch 22-gage (0.0299-inch) steel continuously welded tubing. The front and rear legs shall be pierced to receive the side stretchers and the side stretchers shall be pierced to the receive the cross stretcher at point of junction, and the joining members shall be securely and smoothly welded.

3.3.1.5 *Chafing pad.* — Chafing pads are not required if the rear legs are designed to prevent the chair back from contacting walls. The chafing pad, if furnished, shall be as specified in 3.2.5.

3.3.2 *Type I, style B, chairs.* — The type I, style B chairs shall be of the general design, and not less in size than the minimum dimensions shown in figure 2 and as specified herein. The chairs shall withstand tests as described in 4.3.

3.3.2.1 *Seat.*—The seat shall be the same as specified in 3.3.1.1 except not less than 16 springs shall be used.

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3.3.2.2 *Back.*—The back shall be as specified in 3.3.1.2.

3.3.2.3 *Legs.*—The legs shall be as specified in 3.3.1.4.

3.3.2.4 *Chafing pad.*—The chafing pad shall be as specified in 3.3.1.5.

3.3.3 *Type II, class 1, style A chairs.*—The type II, class 1, style A chairs shall be of the general design, and not less in size than the minimum dimensions shown in figure 3 and as specified herein. The chairs shall withstand the tests described in 4.3.

3.3.3.1 *Seat.*—The seats shall be the same as specified in 3.3.1.1.

3.3.3.2 *Back.*—The back shall be the same as specified in 3.3.1.2 except back legs are not included.

3.3.3.3 *Arms.*—The arms shall be the same as specified in 3.3.1.3.

3.3.3.4 *Base.*—The base shall be fabricated from a one piece die-formed steel sheet, or of die-formed steel sheet sections which shall be fusion welded or brazed into one piece with sufficient strength to meet the test as specified in 4.3.7. The base shall have a plate of not less than 16-gage (0.0598-inch) steel securely welded to the bottom flange of the base and which shall extend out into the legs and shall be accurately pierced for the spindle housing. The base shall have an accurately reamed or burnished spindle housing of not less than 16-gage (0.0598-inch) steel which shall provide a continuous bearing of the spindle for the entire height of the chair base hub. Alternatively, a commercial type insertable hub may be used having self-lubricating bearings. The spindle housing or hub shall be not less than 1-inch-inside diameter and shall have a collar or plate not less than 2½ inches in diameter for securing the housing to the base and a flat smooth bearing surface for the chair control thrust bearing that shall be not less than 3/16 inch wide. The spindle housing shall be securely riveted or welded to the base. The caster mounting shall consist of a 14-gage (0.0747-inch) steel plate and a tube of not less than 16-gage (0.0598-inch) steel to pressure grip the caster socket. The plate shall be the full width of the base leg, and not less than 2⅜ inches long. The tube and plate shall be securely welded together and to the base leg at the top of the tube, and shall securely mount the casters and sockets specified in 3.1.9.1 so that the vertical axis of the casters, when mounted, will be in a true vertical position and parallel with the chair control spindle. The tube may be omitted when plate type casters are used. Scuff plates as specified in 3.1.9.3 shall be securely attached to each base leg so that they cannot become loosened during normal usage. The interior surface of the base shall be coated with sound deadening material or have sound deadening blocks as specified in 3.1.10. Provision shall be made for securing the chair control to the base which will permit revolving without moving the base and permit the removal of the spindle from the housing when desired.

3.3.3.5 *Chair control.*—The chair control shall be as specified in 3.1.9.4 for type II, class 1, styles A and B chairs.

3.3.3.6 *Chafing pad.*—The chafing pad shall be as specified in 3.2.5.

3.3.4 *Type II, class 1, style B chairs.*—The type II, class 1, style B chairs shall be of the general design and not less in size than the minimum dimensions shown in figure 4 and as specified herein. The chairs shall withstand the test described in 4.3.

3.3.4.1 *Seat.*—The seat shall be as specified in 3.3.1.1 except not less than 16 springs shall be used.

3.3.4.2 *Backs.*—The backs shall be the same as specified in 3.3.1.2 except back legs are not included.

3.3.4.3 *Base.*—The base shall be the same as specified in 3.3.3.4.

3.3.4.4 *Chair control.*—The chair control shall be as specified in 3.1.9.4.

3.3.4.5 *Chafing pad.*—The chafing pad shall be as specified in 3.2.5.

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3.3.5 *Type II, class 2, style A chairs.*—The type II, class 2, style A chairs shall be of the general design and not less in size than the minimum dimensions shown in figure 5 and as specified herein. The chairs shall withstand the test described in 4.3 as specified in 3.2.2.2.

3.3.5.1 *Seat.*—The seat shall be the same as specified in 3.3.4.1, except that the upholstery material may be fastened to the underside of the seat frame.

3.3.5.2 *Back.*—The back frame shall be not less than 1-inch by 1-inch square 20-gage (0.0359-inch) steel tubing, and shall be securely attached to the formed cross member of the tilting mechanism. The back rest shall consist of an inner die formed steel pan not less than 20-gage (0.0359-inch) to which the back pad as specified in 3.2.3 shall be secured. The inner pan shall be covered by an outer pan of not less than 20-gage (0.0359-inch) die formed steel. The outer pan shall be given an enamel finish the same as other metal parts of the chair, and shall be secured to the inner pan by not less than four screws or other equally secure method. The back rest shall be flexibly attached to the back frame and shall have a device to permit the position of the back to be adjusted and held to the desired position.

3.3.5.3 *Arm supports.*—The arm support shall be not less than 1-inch by 1-inch 20-gage (0.0359-inch) steel tubing, the lower ends of which shall be securely welded and brazed on all sides to the seat frame. The arm pads shall be as specified in 3.2.4 and as shown in figure 5, or, may not extend down on the arm support if they are over padded and extend beyond the front edge and cover the entire top of the arm. This alternate arm pad shall be secured by not less than two screws with the heads countersunk in the under side of the arm support.

3.3.5.4 *Base.*—The base shall be as specified in 3.3.3.4.

3.3.5.5 *Chair control.*—The chair control shall be as specified in 3.1.9.4 for type II, class 2, styles A and B chairs.

3.3.6 *Type II, class 2, style B chairs.*—The type II, class 2, style B chairs shall be of the general design, and not less in size than the minimum dimensions shown in figure 6 and as specified herein. The chairs shall withstand the test described in 4.3.

3.3.6.1 *Seat.*—The seat shall be the same as specified in 3.3.5.1.

3.3.6.2 *Back.*—The back shall be the same as specified in 3.3.5.2.

3.3.6.3 *Base.*—The base shall be the same as specified in 3.3.3.4.

3.3.6.4 *Chair control.*—The chair control shall be the same as specified in 3.1.9.4 for type II, class 2, styles A or B chairs.

3.4 Finish.

3.4.1 *Cleaning of metal parts.*—All traces of mill scale, rust, oil, dirt and moisture shall be removed before any finishing material is applied. Oil between faying surfaces shall be completely removed to prevent blemishes occurring in the final finish. All welded parts shall have all traces of scale and flux removed before finishing.

3.4.2 *Methods of finishing.*—Chairs shall be given a chemical treatment to provide a nonmetallic phosphate coating, over which there shall be one or more coats of synthetic enamel overtone of the standard color specified, sprayed-on and oven-baked, either convection or infrared, at a temperature and for a period of time as recommended by the enamel manufacturer. The quality, application, drying time, and cleaning shall be controlled to produce a clean, smooth, heavy, even film of finish not less than 1 mil thick, without blisters, pits, wrinkles, runs or tackiness and which shall peel off in ribbons rather than chip, flake or crack when the film is scraped with a knife.

3.5 Product identification.—Unless otherwise specified, each chair shall be marked in a plain, permanent, and inconspicuous manner with the date of manufacture, the manufacturer's name or trade-mark of such known character that the source of manufacture may be readily determined, with the specification number and the Federal Stock Number. When specified, items shall be marked in accordance with Military Standard MIL–STD–130.

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3.6 Workmanship. — The details of construction, including design, joining, assembly, filling, covering, finishing and marking shall be in strict accordance with the detail requirements and tolerances of this specification and the figures. Finished chairs shall be neat, clean and well tailored and shall be free from any defects that affect the appearance or which might affect the serviceability of the chairs or the safety of the users or that might cause injury.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—Normally, the inspection lot size shall consist of all units submitted for inspection at one time.

4.1.2 *Sampling for inspection.* — A representative random sample shall be selected from the lot in accordance with table I. Acceptance or rejection of the lot shall be based on the number of defective units found in the sample selected and inspected, as being equal to or less than the specified acceptance number; or, equal to or greater than the specified rejection number of table I.

TABLE I.—*Sampling for visual and dimensional inspection based on—Acceptable Quality Level (AQL) of 2.5 percent*

Lot size	Sample size	Acceptance number	Rejection number
15 and under	3	0	1
16 to 40	7	0	1
41 to 110	15	1	2
111 to 300	35	2	3
301 to 500	50	3	4
501 and over	75	4	5

NOTE 1.—The Government inspector may institute tightened or reduced inspection in accordance with Standard MIL-STD-105.

NOTE 2.—The above is a single sampling procedure. Alternate double or multiple sampling procedures or alternate Acceptable Quality Levels for different classes of defects may be specified at the discretion of the procuring agency.

4.2 Inspection. — Chairs shall be inspected to determine compliance with this specification, with respect to workmanship, materials, design, details of construction, hardware, finish, tests and marking. Inspection may be made at the factory by a representative of the Government at any time during process of manufacture, or at the point of delivery.

4.3 Test procedures.—Chairs for tests shall be selected in accordance with 4.1.2 from those chairs that have successfully passed the visual inspection called for in 4.2. Points or reference lines shall be provided on various parts of the chair so that accurate measurement may be taken before and after application of test load and with allowance made for any slack in chair control, to determine if any permanent distortion has taken place under test. Any failure of welds or any permanent distortions in excess of that specified, or any other damage to any part of the chairs that would affect the serviceability of the chairs, will be considered as failing to comply with the requirements of this specification.

4.3.1 *Drop test for seats of all types, classes, and styles.* — A bag of sand approximately 16 inches in diameter and of the weight specified in table II for each type, class and style of chair, shall be dropped once onto the center of the seat from a height of 6 inches. The permanent distortion shall not exceed ¼ inch. Dimensions of distortion shall be taken from the floor to the bottom of the seat frame. The drop test on all type II chairs, shall be applied with the seat height set at 1½ inches above the minimum seat height adjustment.

TABLE II.

Type	Class	Style	Test load of
I	..	A	300 pounds
I	..	B	275 pounds
II	1	A	275 pounds
II	1	B	250 pounds
II	2	A	250 pounds
II	2	B	225 pounds

4.3.2 *Back test for fixed back chairs for type I, styles A and B, and type II, class 1, styles A and B chairs.*—With the seat and back cushions removed, the chairs shall be clamped to the floor in an upright position by a bar across the top of the front seat frame, and the legs or base restrained from movement. A spring dynamometer

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or an equally effective device shall be attached to both ends of a yoke fastened to the top arch of the frame and a back pull applied normal to the slope of the back. The chairs shall withstand the load test in pounds specified in table III for each type, class, and style without permanent distortion of more than ¼ inch. The dimensions of distortions shall be taken between the front seat rail and the top of the back frame.

TABLE III.

Type	Class	Style	Test pull of
I	..	A	250 pounds
I	..	B	150 pounds
II	1	A	250 pounds
II	1	B	150 pounds

4.3.3 *Leg test for type I chairs.*—The sample chairs shall be subjected to each of the following leg tests. The dimensions of distortion shall be taken vertically from the floor to the extreme end of the legs.

4.3.3.1 *No. 1 test.*—The chair shall be placed on its side and the lower front leg shall be supported its entire length. Restrain this leg from movement, and apply independently a perpendicular test load one inch from the bottom of each top leg through a ½-inch-diameter rod. The chairs shall withstand a load of 125 pounds before a permanent distortion of ¼ inch occurs.

4.3.3.2 *No. 2 test.*—The chair shall be placed on its back. Restrain the lower leg from movement. A perpendicular test load shall be applied independently through a ½-inch-diameter rod placed perpendicular to the long axis of each front leg and 1 inch from the bottom of the legs. The chair shall withstand a test load of 250 pounds before a permanent distortion of ¼ inch occurs.

4.3.4 *Racking test for all leg chairs.*—Remove the seat and back cushions. The chair shall be tilted on its rear legs, and the legs restrained from movement. A greased metal faced beam shall be placed at top of the front center of the seat frame and at the center top edge of the back frame. Apply a downward test load as specified below, at a point vertically above the intersection point of the rear legs and the floor. The dimension for distortion shall be taken between the front seat rail or frame and the top of the back frame. Type I, style A chairs shall withstand a test load of 600 pounds before a permanent distortion of ⅛ inch (maximum) occurs. Type I, style B chairs shall withstand a test load of 400 pounds before a permanent distortion of ⅛ inch (maximum) occurs.

4.3.5 *Arm test for type I, style A, and type II, class 1, style A chairs.*—With the chair clamped to the floor in an upright position, and the arm pad and seat cushion removed, restrain the seat frame but not the arm from movement. A spring dynamometer or an equally effective device shall be attached to the outer top front end of the arm and a horizontal pull shall be applied in an outward direction perpendicular to the plane of the arm. The arm shall withstand a test pull of 175 pounds before a permanent distortion of ¼ inch occurs. The dimension of distortion shall be taken horizontally to the top of the arm frame.

4.3.6 *Normal test for type II, classes 1 and 2, styles A and B chairs.*—Restrain the base of the chair from movement. The seat height shall be set 1½ inches above the minimum seat height adjustment and the back tension adjusted to the mean or normal tension position. On all posture chairs, the back height and horizontal adjustments shall be set at the medium position. A spring dynamometer or an equally effective device shall be attached to both ends of a yoke fastened to the top of the back frame, and a back pull shall be applied normal to the slope of the back at all times and up to the maximum limit of travel. The chairs shall withstand load test specified in table IV for each type, class, and style without a permanent distortion of more than ¼ inch. Measurements for determining distortion shall be taken horizontal to the top of the back frame.

TABLE IV.

Type	Class	Style	Test pull of
II	1	A	200 pounds
II	1	B	175 pounds
II	2	A	150 pounds
II	2	B	100 pounds

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4.3.7 *Base test.*—With the casters removed and the base only in an upright position supported only at the usual point of caster contact with steel blocks of sufficient height to prevent center tube from touching testing platform, apply a static load of not less than 2,500 pounds at the rate of ⅛ inch per minute and concentrated over an area of 1¾ inches in diameter at the top center of the base. The base shall withstand the load without any permanent distortions, fractures, or interior or exterior breakage.

4.3.8 *Cotton felt.*—The cotton felt shall be examined before use for compliance with 3.1.3 by classifiers licensed by the U. S. Government Department of Agriculture, or, representative samples, each weighing not less than ½ pound, may be forwarded to the Board of Cotton Examiners, U. S. Department of Agriculture, Washington 25, D. C. The opinion of this board as to the true grade or quality shall be final. Approvals, properly identified, shall be made available to the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 Application.—Subject commodity shall be packaged and packed for domestic shipment (immediate use), domestic shipment (storage), or overseas shipment (commercial export), or overseas shipment (military), as specified. (See 6.1.)

5.2 Domestic shipment (immediate use).—Unless otherwise specified, subject commodity shall be packed in containers, which (together with packaging, wrapping and inner packing) are acceptable by common or other carriers for safe transportation to point of detination specified in shipping instructions at the lowest transportation rate for such supplies.

5.3 For Domestic shipment (storage).—When specified.

5.3.1 *Packaging.*

5.3.1.1 *Type I, style A or B.*—Each chair shall have all finished surfaces (as applicable) of legs, arms, rails, and backrest completely covered with cushioning material securely tied by a spiral winding of jute twine having a breaking strength of not less than 30 pounds. The spiral windings shall not be more than 2 inches apart. Cushioning material may be macerated paper pads, wood wool packing pads, wood fibre felt pads, or creped cellulose wadding pads. All seats shall be completely covered with a sheet of kraft paper having a minimum basis weight of 60 pounds per ream (24 x 36—500), secured with cross ties of twine.

5.3.1.2 *Type II, classes 1 and 2, styles A and B.*—Each chair, set and secured at its lowest rotary position and with casters removed, shall have all surfaces of the legs, arms and backrest (as applicable) cushioned as specified in 5.3.1.1. The chair seat shall be completely covered with kraft paper as specified in 5.3.1.1. The casters shall be placed in a good quality cloth bag with drawstring top and fastened to the rotary mechanism of the chair in a manner that precludes disengaging in handling or transit.

5.3.2 *Packing.*

5.3.2.1 *Type I, style A or B.*—Two chairs shall be packed together into a corrugated fiberboard container. Invert one chair over a second chair bringing the seats together. Chairs with arms shall be inverted and arms staggered into position, bringing the arms to rest on the seats of the chairs. Tightly tie front leg and back leg post of each chair together with twine having a breaking strength of not less than 80 pounds and lock knot securely. Boxes shall conform to the requirements of Federal Specification LLL-B-631, style FOL, except that the length of the outer flaps may be not less than the inside width of the box minus 1. Each box shall be fabricated from double faced corrugated fiberboard, having a minimum bursting strength of 275 p.s.i. All flaps of the fiberboard box shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps. All seams shall be sealed by the application of 3-inch wide gummed paper tape complying with Federal Specification UU-T-111, class 2.

Alternatively, shipping container may be style RSC. When a style RSC container is used it shall be provided with a four-piece wood frame top and bottom as a support for the chairs. The frames shall be not less than ⅝ inch in thickness and of sufficient width to prevent the chair legs

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from shifting off frames in transit. Frames shall be clinch-nailed at each point of attachment. Chair legs shall not be attached to frame.

5.3.2.2 *Type II, classes 1 and 2, styles A and B.*—One chair shall be packed for shipment in a corrugated fiberboard box. Boxes shall be fabricated and sealed as specified in 5.3.2.1 except that style RSC shipping container shall not be used.

5.4 For overseas shipment (commercial export). — When specified, subject commodity shall be packed in containers which, together with packaging wrapping and inner packing, are acceptable by ocean-going vessels for safe transportation to the points of destination specified in the shipping instructions at the lowest transportation rate for such supplies.

5.5 For overseas shipment (Military).— When specified.

5.5.1 *Packaging.* — Packaging shall conform with paragraph 5.3.1.

5.5.2 *Packing.* — Two chairs of the type I, style A or B, positioned and secured as specified in paragraph 5.3.2.1, or one chair of the type II, classes 1 and 2, styles A and B, shall be packed in a nailed wood box. The nailed wood box shall be constructed and strapped in accordance with Military Specification JAN-P-106, style 2, 2½, or 4. Each shipping container shall be provided with a case liner conforming with Military Specification MIL-L-10547, types I or II, grades B or C, class 2.

5.6 Marking.—Shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

5.6.1 *Special marking.*—The top panel shall be marked as follows:

FURNITURE
FRAGILE, HANDLE WITH CARE
TOP
THIS SIDE UP

In addition, one front panel shall be marked in characters not less than two inches high as follows:

UP FURNITURE UP
FRAGILE, HANDLE WITH CARE

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein and should specify the type, class and style of chair required, the color of finish required and whether chairs should be packaged and packed for other than domestic shipment, immediate use (see 1.2.1, 1.2.2, 3.1.5.2, 3.2.2.2, 3.2.3, 3.5, 4.1.2 note 2, 5.1 and 5.6.)

(a) Domestic shipment (immediate use), applies to that material required for immediate use without storage or reshipment by the initial domestic receiving activity (see 5.2).

(b) Domestic shipment (storage), applies to that material for storage and reshipment to a domestic receiving activity not involving sea transportation (see 5.3).

(c) Overseas shipment (commercial), applies to that material which is transported by ocean-going vessel, accorded normal covered storage at destination and where the end use of the item is to be at the initial receiving activity (see 5.4).

(d) Overseas shipment (military), applies to that material to be transported by ocean going or coastwise vessel and handled, issued, and stored in the military supply system (see 5.5).

6.2 Federal specifications do not include all the types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc. which are suitable for Federal Government requirements and are as referenced in Federal Standard No. 93.

Patent notice.—When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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6.3 Transportation description.—Transportation description applicable to this item is:

Type I

Chairs, not otherwise indexed by name, steel or with steel or wire frames. (Indicate set-up or knocked down).

Carload minimum weights:

In cars 40′ 9″ or less in length, inside measurements—10,000; 12,000; 15,000; 18,000; 22,000; 30,000 pounds.

In cars exceeding 40′ 9″ in length, inside measurements—16,000; 18,000; 21,000; 24,000; 30,000; 40,000 pounds.

Truckload minimum weights:

S.U.—10,000 pounds.

Backs removed and bases nested, 16,000 pounds.

Backs and seats separated and removed from frames, frames interlaced or K.D., other than backs removed and bases nested, 24,000 pounds.

Type II

Chairs, metallic, revolving, not otherwise indexed by name. (Indicate set-up or knocked-down).

Carload minimum weights:

In cars 40′ 9″ or less in length, inside measurements—10,000; 12,000, 15,000; 18,000; 22,000; 30,000 pounds.

In cars exceeding 40′ 9″ in length, inside measurements—16,000; 18,000; 21,000; 24,000; 30,000; 40,000 pounds.

Truckload minimum weights:

S.U.—10,000 pounds.

K.D.—16,000 pounds.

Maximum weights shown apply when material is transported at classification ratings. They will vary when material moves on commodity rates. When in doubt consult General Services Administration Regional Traffic Manager.

MILITARY INTEREST:

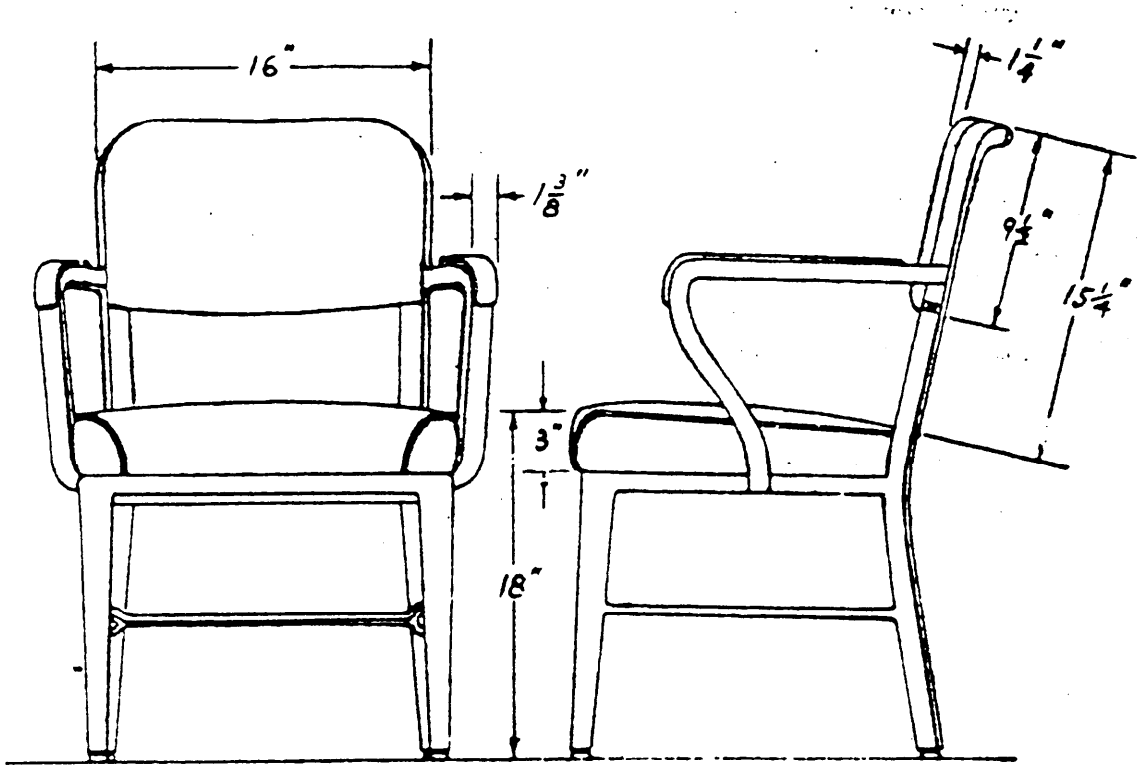
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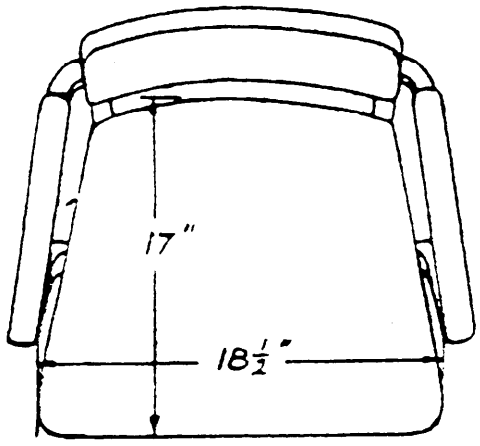
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Air Force.

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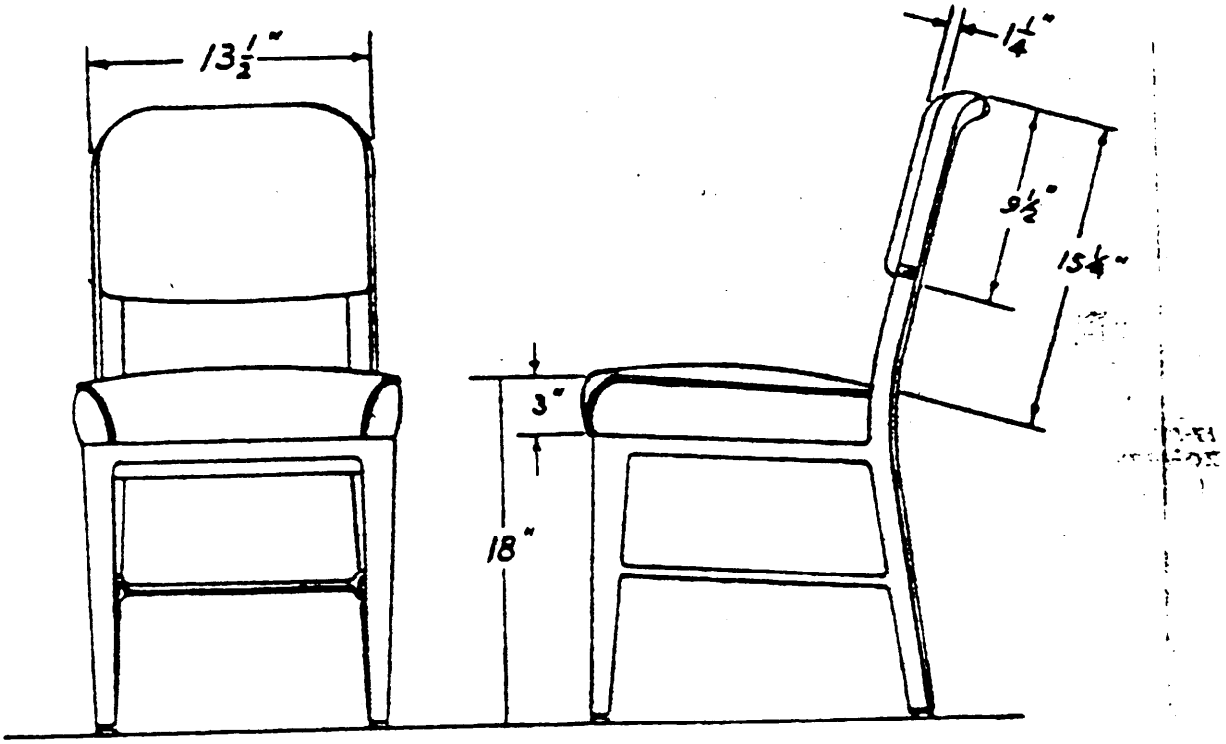


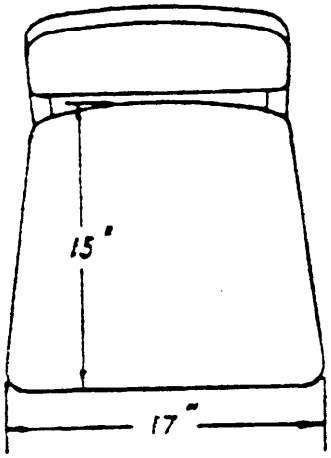
TYPE I
STYLE A

MINIMUM DIMENSIONS

FIGURE 1

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TYPE I
STYLE B

MINIMUM DIMENSIONS

FIGURE 2.

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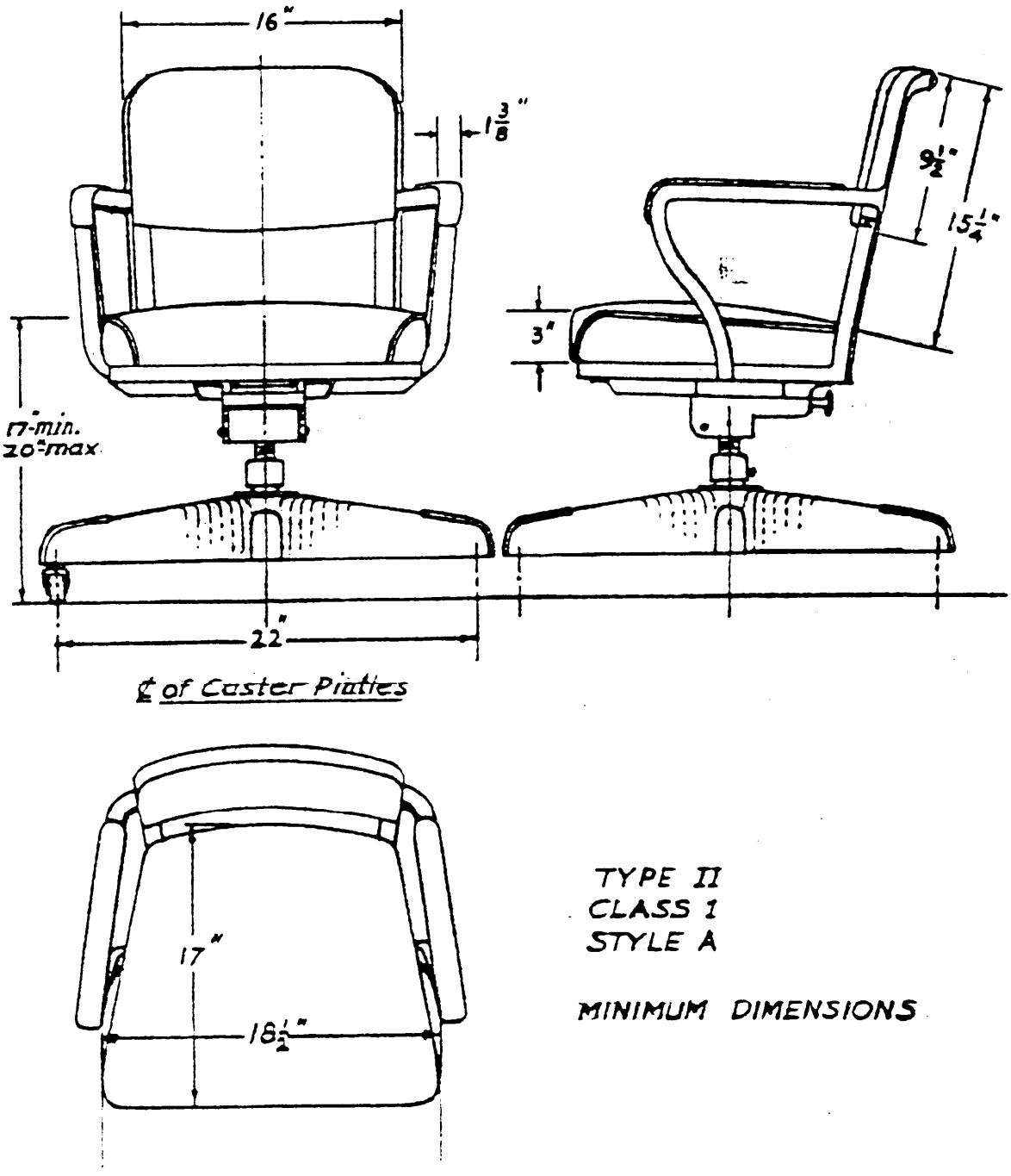
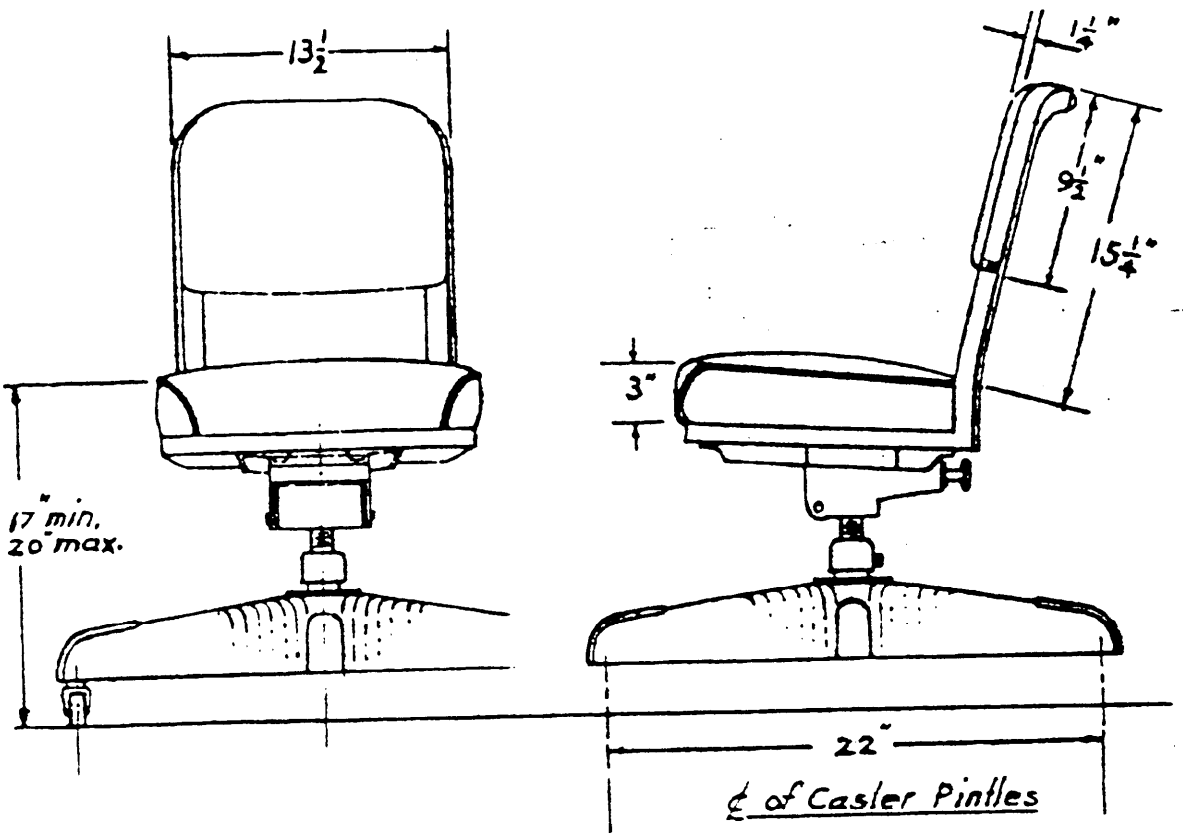
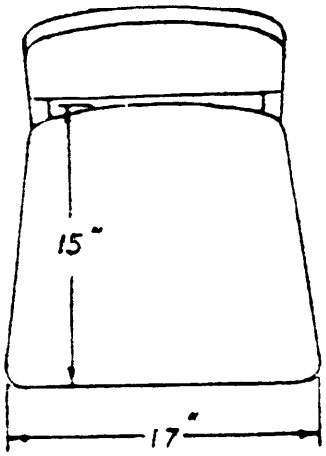


FIGURE 3.

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TYPE II
CLASS 1
STYLE B

MINIMUM DIMENSIONS

FIGURE 4.

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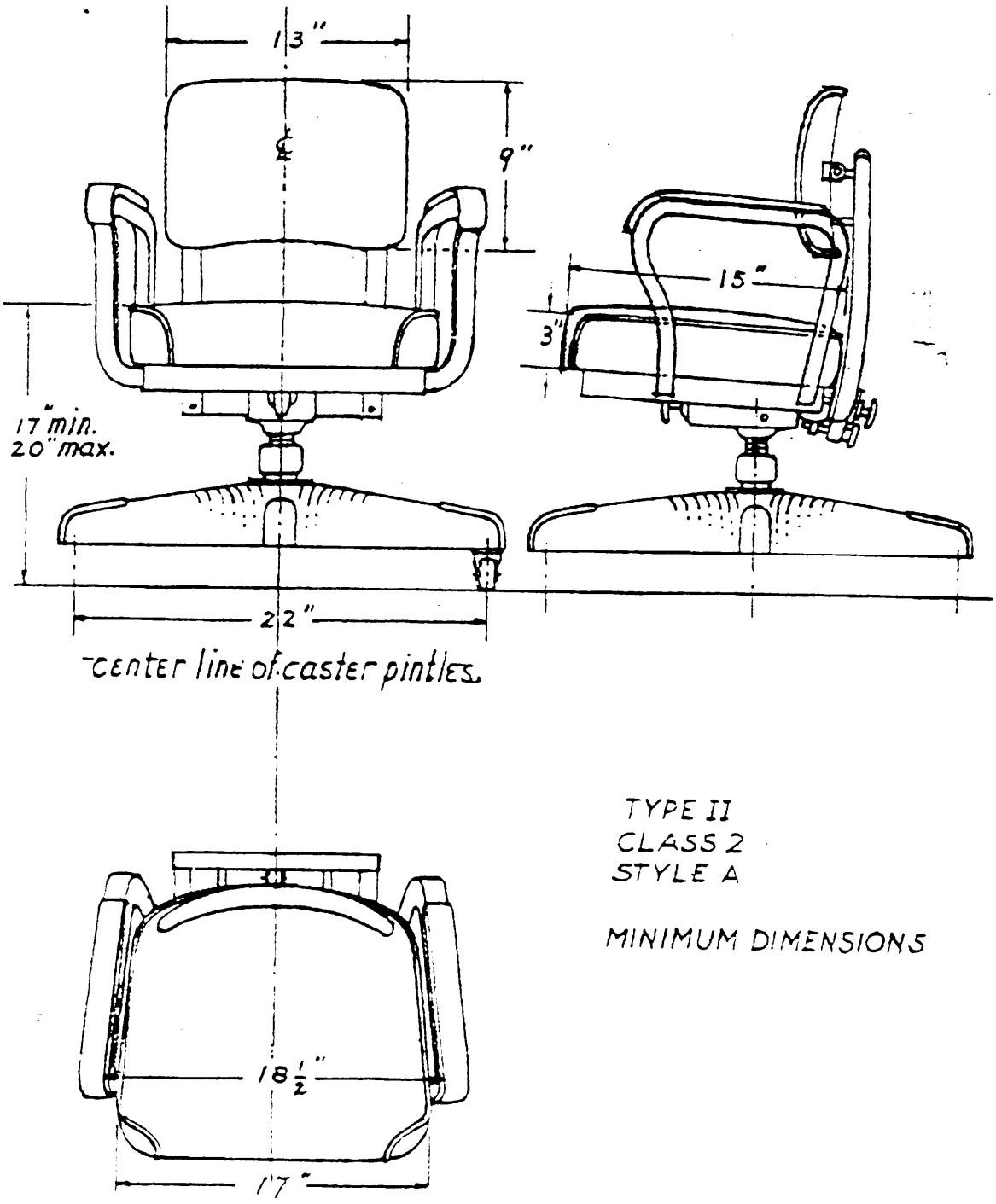
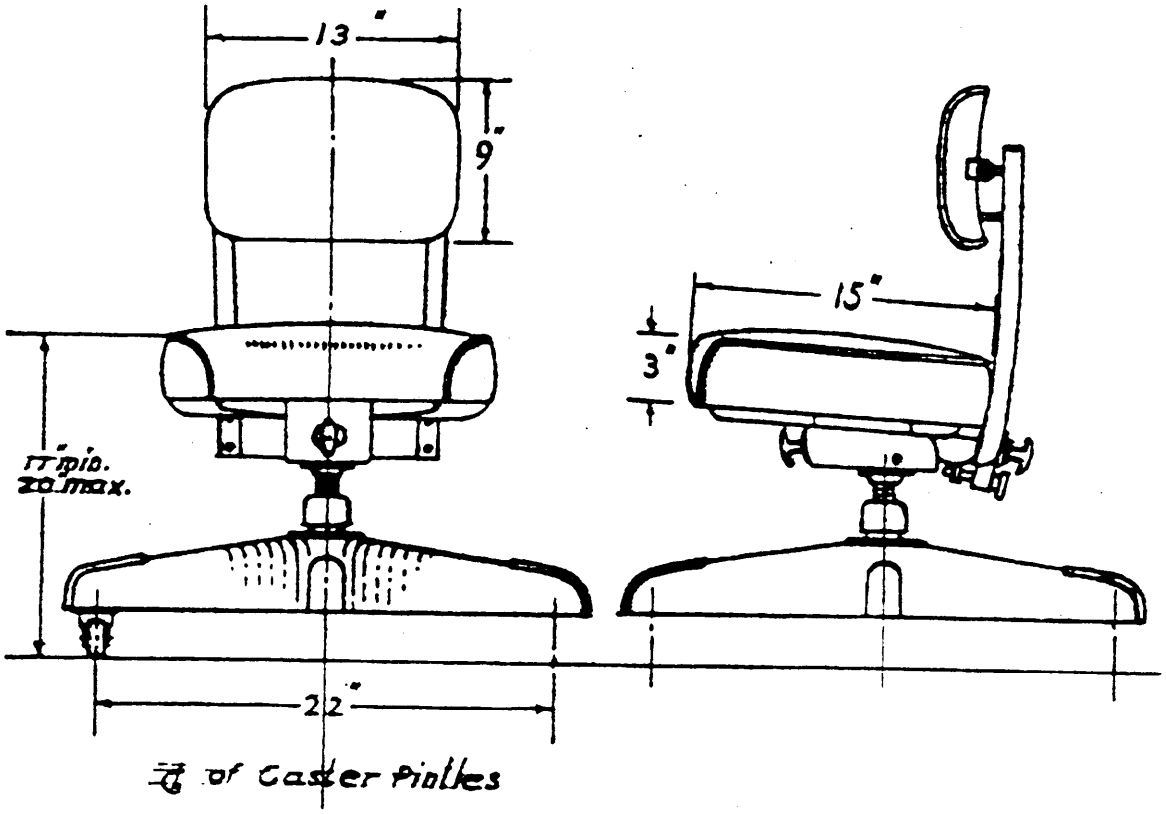
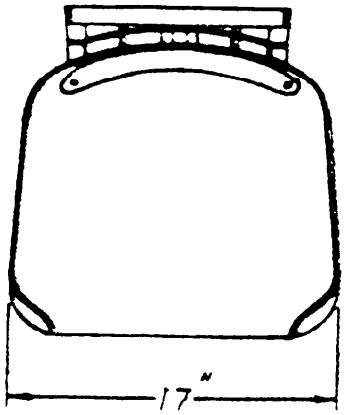


FIGURE 5.

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TYPE II
CLASS 2
STYLE B

MINIMUM DIMENSIONS

FIGURE 6.

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